

University of Puerto Rico Mayagüez Campus
Department of Electrical and Computer Engineering
Project Assignment 7 CSP Programming Assignment:
Simplified Zebra Problem

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Abstract:

This report analyzes the Zebra logic puzzle as a Constraint Satisfaction Problem (CSP). The puzzle involves five houses, five colors, five nationalities, five candy brands, five drinks, and five pets. Different CSP representations are discussed, including a house-based representation, an attribute-position representation, and a binary constraint representation. The analysis shows that the zebra belongs to the Japanese, and water is drunk in the first house by the Norwegian. Among the possible CSP representations, the attribute-position model is usually preferred because it is compact, clear, and efficient for constraint propagation.

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Introduction:

Constraint Satisfaction Problems (CSPs) are a class of artificial intelligence problems in which a solution is obtained by assigning values to a set of variables while satisfying a collection of constraints. In a CSP, each variable has a domain of possible values, and the objective is to determine a complete and consistent assignment that fulfills all the imposed restrictions. CSPs are widely used in artificial intelligence because many real-world problems can be represented naturally using variables, domains, and constraints. Examples include scheduling, map coloring, cryptarithmic puzzles, transportation planning, and resource allocation problems.

The objective of this programming assignment is to formulate and solve the simplified Zebra Problem as a Constraint Satisfaction Problem. The Zebra Problem is a well-known logic puzzle involving five houses, each with unique attributes such as color, nationality, candy preference, drink, and pet ownership. A collection of logical clues constrains the possible combinations of attributes. The goal is to determine where the zebra lives and which house drinks water by systematically applying the constraints provided in the problem statement.

This assignment also explores different possible representations of the Zebra Problem as a CSP. Different formulations may affect the complexity of the search process, the efficiency of constraint propagation, and the readability of the implementation. The report discusses alternative representations and analyzes the advantages and disadvantages of each approach. Particular attention is given to the relationship between CSP structure and computational efficiency.

The solution developed for this assignment uses a CSP representation based on variables, domains, and binary constraints together with a backtracking search strategy. Constraint relationships are enforced incrementally during the search process to reduce invalid assignments and improve computational efficiency. The implementation demonstrates how classical CSP techniques can be applied to solve structured logical reasoning problems effectively.

Theory:

Constraint Satisfaction Problems (CSPs) are a category of artificial intelligence problems in which the objective is to assign values to a set of variables while satisfying a collection of constraints. Unlike standard search problems, where states are treated as black boxes, CSPs explicitly define variables and the domains of values that each variable may assume. A solution is obtained when all variables are assigned values that satisfy every constraint in the problem.

A CSP is formally represented using three primary components: variables, domains, and constraints. Variables represent the unknown elements of the problem, domains define the possible values available to each variable, and constraints restrict the combinations of values that variables may simultaneously assume. Constraints may be unary, binary, or higher-order depending on the number of variables involved.

One common representation used in CSPs is the constraint graph, where nodes represent variables and edges represent constraints between variables. This representation helps visualize dependencies among variables and can reduce the search space more efficiently. CSPs are often solved using techniques such as backtracking search and constraint propagation, including node consistency and arc consistency, which eliminate invalid assignments before a complete solution is generated.

Constraint Satisfaction Problems are widely used in artificial intelligence because they provide an efficient framework for solving combinatorial problems such as scheduling, map coloring, transportation planning, cryptarithmic puzzles, and logic problems. By exploiting the structure of constraints, CSP methods can significantly reduce computational complexity when compared to brute-force search methods.

CSP Representations of the Zebra Problem

3.1 House-Based Representation

In this representation, each house is a variable. Each house contains a full tuple:

$\text{House}_i = (\text{color}, \text{nationality}, \text{candy}, \text{drink}, \text{pet})$

The domain of each house is all possible combinations of attributes. Constraints are then applied to eliminate invalid combinations.

This representation is intuitive because it follows the natural wording of the puzzle. However, it can become inefficient because each house has many possible combinations.

3.2 Attribute-Position Representation

In this representation, each attribute is treated as a variable, and the value is the house position.

Example:

Englishman = 3

means the Englishman lives in house 3.

Variables include:

- Englishman, Spaniard, Norwegian, Ukrainian, Japanese
- Red, Green, Ivory, Yellow, Blue
- Hershey, Kit Kats, Smarties, Snickers, Milky Ways
- Coffee, Tea, Milk, Orange Juice, Water
- Dog, Fox, Snails, Horse, Zebra

Each variable has the domain:

{1, 2, 3, 4, 5}

The constraints include:

- Englishman = Red
- Spaniard = Dog
- Norwegian = 1
- Green = Ivory + 1
- Hershey is next to Fox
- Kit Kats = Yellow
- Norwegian is next to Blue
- Smarties = Snails
- Snickers = Orange Juice
- Ukrainian = Tea
- Japanese = Milky Ways
- Kit Kats is next to Horse
- Coffee = Green
- Milk = 3

This representation is the simplest because the constraints become equations or position relationships. This representation simplifies the problem because many clues can be expressed directly as equality, inequality, or positional constraints between variables.

3.3 Binary Constraint Representation

Another possible representation is to create binary constraints between pairs of variables. For example:

Englishman = Red

is a binary equality constraint, while:

$| \text{Hershey} - \text{Fox} | = 1$

is a binary adjacency constraint.

This representation works well with standard CSP algorithms like backtracking, forward checking, and arc consistency. However, it may require many all-different constraints to ensure that each color, nationality, candy preference, drink, and pet appears exactly once across all houses. These uniqueness constraints are essential for maintaining consistency throughout the CSP formulation.

Comparison of Representations

The house-based representation is easiest to understand visually because each house is treated as one complete unit. However, it is less efficient because the solver has to consider many possible combinations for each house.

The attribute-position representation is more simple. It directly represents each person, color, candy, drink, and pet as a variable whose value is a house number. This makes clues easier to translate into constraints.

The binary constraint representation is useful for algorithms because it breaks the problem into small relationships between variables. However, it may be less readable for humans because the complete house arrangement is not shown directly.

Overall, the attribute-position representation provides the clearest and most efficient formulation of the Zebra Problem. By representing each attribute as a variable associated with a house position, most clues can be translated directly into equality, inequality, or adjacency constraints. This representation simplifies the reasoning process and is well suited for classical CSP techniques.

Results

House	Color	Nationality	Candy	Drink	Pet
1	Yellow	Norwegian	Kit Kats	Water	Fox
2	Blue	Ukrainian	Hershey Bars	Tea	Horse
3	Red	Englishman	Smarties	Milk	Snails
4	Ivory	Spaniard	Snickers	Orange Juice	Dog
5	Green	Japanese	Milky Ways	Coffee	Zebra

As can be seen, the zebra lives in the rightmost house, owned by the Japanese, and the water is in the first house on the left.

Conclusion

The Zebra puzzle can be represented as a CSP in several ways. A house-based representation is intuitive but computationally less efficient because each variable contains many possible attribute combinations. A binary constraint representation is useful for formal CSP algorithms but can become lengthy. The attribute-position representation is the most straightforward in pen and paper because it expresses the puzzle's clues clearly and allows constraints such as equality, adjacency, and immediate-right relationships to be modeled directly.

The solution obtained from the CSP is that the Japanese own the zebra, and the Norwegian drinks water.

References

[1] N. G. Santiago and M. A. Jiménez, *Writing Formal Reports: An Approach for Engineering Students in 21st Century*, 3rd ed., University of Puerto Rico at Mayagüez, 2002.

[2] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 3rd ed. Pearson, 2010.